FINANCING CHOICES OF ENTREPRENEURS IN THE VENTURE LIFECYCLE: PATTERNS AND CHALLENGES IN A DEVELOPING ECONOMY

INTRODUCTION

Small and medium sized enterprises (SMEs) need access to financial capital to fund their operations, acquire technology and pursue growth opportunities (Acs, 2010). Entrepreneurs then choose the appropriate mix of finance sources (e.g., debt, equity, retained earnings) to fund their ventures. This choice, with long-term consequences for the viability of the venture, is influenced by individual-level, firm-level and institutional-level factors (Nguyen & Canh, 2021). Bergen and Udell (2006), for instance, find that small firms tend to rely on personal loan guarantees and the owners' funds whereas larger firms choose a mix of debt and equity financing. Institutional level factors affecting SMEs' financing choices are perhaps most salient in emerging countries where SMEs must navigate poorly-developed capital markets characterized by high transaction and capital costs (Khanna et al., 2005).

Women entrepreneurs account for a significant portion of SME founders (Elam et al., 2019). As such, a substantial stream of research within the entrepreneurship literature has debated the existence and sources of systematic differences in access to financing choice between female and male entrepreneurs. On one hand, some scholars argue that there are no systematic differences between men and women entrepreneurs. Fabowale et al. (1995), using a sample of 2,763 SMEs in Canada, find no differences in credit terms offered to male and female business owners after controlling for differences in the attributes of firms owned by entrepreneurs.

On the other hand, research suggests there are indeed systematic differences in financing choice between male and female entrepreneurs. These may be due to differences in individual preferences as women entrepreneurs rather personal savings and financing from friends and family than external financing sources (Coleman, 2000), and gender-based discrimination as women are treated unequally in credit decisions (Berger & Udell, 2006; Chaudhuri et al., 2020). The nature of financing source has also been implicated in these differences between male and female business owners. Women entrepreneurs are less likely to receive bank financing (Eddleston et al., 2016), private equity funding (Becker-Blease & Sohl, 2007), venture capital financing (Canning et al., 2012), but equally likely to obtain equity capital (Orser et al., 2006), and may even have better access to crowdfunding than their male counterparts (Johnson et al., 2018). Verheul and Thurik (2001) suggest that the reasons for any differences may be manifold: indirect effects due to differences in the size of business, type of business, quality of management between men and women business owners, as well as direct effects due to discriminatory attitudes of funders.

We contribute to this debate by asking the following research questions: (1) Do male and female small business owners exercise different financial choice at various stages of a venture's lifecycle, i.e., inception, early growth and maturity? (2) If so, what factors influence those choices at the various stages of the venture lifecycle?

THEORETICAL BACKGROUND

Pecking order theory (Myers, 1984; Myers & Majluf, 1984) explains how firms choose their capital structure. Firms prefer to fund investments using internal sources of finance (retained earnings) rather than external sources, such as debt or equity. If firms require external funding, then they prefer to use debt financing over equity financing. The main drivers for the preference order are (1) *Maintaining control*—Managers prefer internal sources of funding that do not require them to concede ownership or control. (2) *Avoiding monitoring costs*—To reduce information asymmetry, external investors are likely to monitor the firm as a condition for providing finance. Managers, keen to avoid the costs of monitoring, privilege internal funds (with no monitoring costs) over external sources (with attendant monitoring costs).

Pecking order theory was originally developed to explain the capital structure of large, public corporations, characterized by separation of ownership and control, transferrable shares, and perpetual life (Blair & Stout, 1999). As firms transition from early stage start-up to growth, their financing needs are likely to change as their internal organization and external environment change (Churchill & Lewis, 1983; Fisher et al., 2016; Winborg & Landström, 2001). Thus, scholars applying pecking order theory to entrepreneurial ventures suggest that entrepreneurs maintain a hierarchy of preference with internal funds as most preferred followed by bank loans and equity finance (Cassar, 2001). Unlike managers of large publicly-traded firms, however, SME owners often do not know about the availability of financing options and endure higher costs of capital (Holmes and Kent, 1991). This “small business gap” (Holmes & Kent, 1991) means that small firms have limited options to issue equity and their owners are strongly averse to dilution of ownership. Hence, Holmes and Kent (1991) propose a constrained pecking order for SMEs, whereby small firm owners prefer their own equity as initial capital source, followed by retained earnings, and finally, short-term debt in the form of bank overdrafts and/or trade credit.

The insights from constrained pecking order model presuppose relatively well functioning (and accessible) debt and capital markets. We use Holmes and Kent’s (1991) constrained pecking order framework to address our questions in the context of Kazakhstan, the largest economy in Central Asia.

METHODS

Research Context

Since gaining independence from the Soviet Union in 1991, Kazakhstan has modelled itself after Singapore as a developmental state (Michel, 2015). The government has reformed the law and promoted policies to stimulate private enterprise as part of its long-term development agenda (Filimonau et al., 2022). It has particularly targeted entrepreneurship among women. Through its enterprise development fund DAMU, for instance, the government provides preferential loans and training to women entrepreneurs (Asian Development Bank, 2013). Despite these advances, Kazakhstan’s entrepreneurs operate in a nascent market economy characterized by low competition and anti-competitive barriers. The economy remains dominated by a few large state-owned firms: 20% of firms in Kazakhstan contribute 90% of value added in the economy (Konings et al., 2022). Moreover, SMEs have to contend with weak or arbitrary regulatory regimes overrun by vested interests (especially in the financial sector), costly and ineffective trade agreements, and corruption (Bohr et al., 2019).

Research Design

We adopt a mixed-methods approach, combining quantitative and qualitative analyses in two studies (hereafter Study 1 and Study 2) to address the research questions. We describe both studies below.

Study 1. We surveyed 1,500 SMEs randomly selected on a sample frame of 80,000 active firms registered with Kazakhstan’s Chamber of Commerce. Survey questionnaires were originally developed in English in 2018 and then translated into Kazakh and Russian—Kazakhstan’s official languages. After translation, the questionnaires were pre-tested with 50 native Russian and Kazakh speakers. These included MBA students and local entrepreneurs. Based on feedback from the pilot, the survey was emended to clarify the questions and more carefully obtain culturally-sensitive information. The survey questions were administered in the second half of 2019 with the selected entrepreneurs over telephone by a reputable social science research agency. The survey covered detailed information about the entrepreneur’s sources of financing, as well as measures of financial literacy. In addition, the survey included data on characteristics of each enterprise such as location, years of operations, sector, employment and sales, among others.

Study 2. We conduced in-depth semi-structured interviews with 14 female business owners in two cities, Kokshetau and the capital city, Astana. Interviews with women entrepreneurs in Astana, the capital city of Kazakhstan, are ongoing. We asked about the sources of finance for investment and working capital at various stages in their business, reasons for their choice, their preferences, and their perceptions of barriers to access to finance. We included an open-ended section where we asked for respondents’ views about obtaining finance. Data collection is going. So far, we do not have a comparative sample of male entrepreneurs. The qualitative data is intended towards a deeper understanding of the women entrepreneurs’ financing choices and the factors that influenced their decisions.

RESULTS

Table 1 presents descriptive statistics on (1) firm characteristics; (2) individual characteristics of the small business owners; and (3) their financing choices at the start-up stage, for capital expenditure, and for working capital. 45% (676) of the 1,500 firms in our sample are represented by women and 55% (824) by men. Women-led firms are older, but employ fewer full-time and part-time employees. Furthermore, as the last six rows of Table 1 indicate, 50% of women-led firms are engaged in wholesale and retail trade and least likely to be involved in the transportation and storage industry (3%). Male-led firms are more evenly spread out, with the highest proportion in wholesale and retail (28%) and in agriculture, forestry and fisheries (25%). Male and female-led SMEs are equally likely to operate in the manufacturing industry (4-5%).

Financial literacy was computed as an index (0 – minimum value; 4 – maximum value) based on participants' responses to four questions on compound interest, real interest rates, risk diversification, and limited liability regulation. Male entrepreneurs have significantly higher financial literacy than their female counterparts and are more likely to use computers in their businesses while female entrepreneurs are more risk averse. There is no significant difference in educational attainment between men and women.

As for financing preferences at the business start-up phase, capital expenditure for expansion, and working capital when the firm is a going concern, male and female entrepreneurs raise funds predominantly through self-financing. Apart from the most-preferred reported avenues for financing capital (Table 1), we identified other financing sources such as trade credit, loans from family and friends, non-bank sources, and equity. At the business start-up phase, while both male and female entrepreneurs show similar preference for raising capital through self-financing, women entrepreneurs depend less on government funding or other development funding and more on local commercial banks for sourcing their capital compared to male entrepreneurs.

Next, about 35 per cent of women entrepreneurs invested in capital expenditure in the financial year (FY) at the time of the survey, statistically significantly less than male entrepreneurs (43 per cent invested in capex in preceding the survey. Of those who invested in capital expenditure, both male and female entrepreneurs were equally likely to seek financing from either local commercial banks or from government or development funds. However, although 50 per cent of female entrepreneurs who invested in capex self-financed the funds, significantly more male entrepreneurs (61 per cent) preferred own funds for capex.

Finally, a significantly lower proportion of female entrepreneurs (38 percent) reported having invested in working capital for the firm in the last FY at the time of the survey compared to the male entrepreneurs (43 per cent). Of those who invested in the working capital, compared to the male entrepreneurs, a greater proportion of female entrepreneurs raised funds via local commercial banks, lesser per cent females depended on government funding, and an equal proportion of women entrepreneurs self-financed their working capital expenditures. (All the above stated results are statistically significant at the five per cent level.)

To understand better patterns of financing behaviour at various stages of raising capital, we turn to findings from Study 2. We present preliminary findings from thematic analysis of the qualitative data (Braun & Clarke, 2006).

Table 1. Descriptive Statistics

Variables	Gender of the entrepreneur		*p*-value of difference (t-test)
	Female	Male	
Firm age - Number of years the firm has been in operation	11.68	10.33	0.00
Number of full-time employees	3.30	4.38	0.05
Number of part-time employees	1.12	2.84	0.01
Financial literacy (Mean scores)	2.52	2.63	0.03
Proportion of entrepreneurs (within gender) who -			
Use laptops/computers for business purposes (measure of business sophistication)	0.47	0.55	0.00
Are risk averse	0.31	0.20	0.00
Possess educational attainment of bachelor's degree and higher	0.52	0.51	0.58
Depend on entrepreneurship as main source of income	0.70	0.73	0.05
Raised initial capital from commercial banks	0.18	0.14	0.04
Raised initial capital from government funding/other development funding sources	0.05	0.09	0.00
Raised initial capital through self-financing	0.60	0.62	0.18
Invested in capital expenditure in previous financial year (FY)	0.35	0.43	0.00
Invested in capital expenditure by raising funds from commercial banks	0.25	0.17	0.18
Invested in capital expenditure by raising funds from government funding/other development funding sources	0.07	0.11	0.07

Invested in capital expenditure through self-financing	0.50	0.62	0.01
Invested in working capital in previous financial year (FY)	0.38	0.43	0.03
Invested in working capital by raising funds from commercial banks	0.17	0.11	0.04
Invested in working capital by raising funds from government funding/other development funding sources	0.03	0.09	0.00
Invested in working capital by raising funds through self-financing	0.60	0.68	0.06
Applied for business loan to a bank in the last three years (at least once)	0.36	0.38	0.13
Are engaged in - Wholesale and retail trade; car and motorcycle repair industry	0.50	0.28	0.00
Are engaged in - Agriculture, forestry and fisheries industry	0.07	0.25	0.00
Are engaged in - Real estate operations	0.10	0.05	0.00
Are engaged in - Transport and storage industry	0.03	0.12	0.00
Are engaged in - Accommodation and Food Services industry	0.06	0.04	0.02
Are engaged in - Manufacturing industry	0.04	0.05	0.29

The size of these 14 firms, whose founders we interviewed, ranged from three to 30 employees. Ten of the firms were started after 2010. For some female entrepreneurs in the sample, financing constraints and risk aversion determined the sector they in which they established their ventures. As Melissa, an education coach, recalled, "I was thinking how I would do such business (of educational coaching). At that time, it seemed impossible, I couldn't wrap my head around the whole idea. That's why I chose the niche where I don't have to invest anything." We list the themes emerging from the thematic analysis of data.

Theme 1. Preference for Self-finance

Most women entrepreneurs preferred bootstrapping and self-finance at early stages of their venture. Their preferences were driven by unavailability of alternative sources of finance—"I had no other source. I didn't even think about credit," Linda, one of the women interviewed observed—and onerous collateral demands. Gera recalled, "I wanted to take out a loan at 6% per annum in second-tier [commercial] banks, there was a condition that I have a high turnover. Besides, additional collateral was needed, which at that time I didn't have."

The main types of bootstrapping at venture start-up stage include use of personal savings, in-kind support from friends and family, zero-interest rate loans from friends, family members seeking personal loans from the banks on entrepreneurs' behalf. Mieke, a small business owner recalled how she raised money to start her venture, "No, I was alone, but my husband always supported me…When I opened my first store, my husband's brother took out a loan for 1.5 million tenge [$3,500]. They (the bank) didn't approve my loan."

We also observed that some female entrepreneurs relied on subsidized loans and business loans from banks, and loans from microfinance organizations.

11 of the 14 women entrepreneurs interviewed depended primarily on their personal savings or retained earnings from their ventures to finance capital expenditure and working capital in their ventures. They were deterred by high interest rates from taking business loans from banks. (They often took personal consumer loans.) We observed that a minority of women accessed grants and government-subsidized loans targeted at female entrepreneurs. While interest rates on these loans were lower than on business loans, the monitoring and reporting requirements set by the Kazakhstan government deterred female entrepreneurs from accessing the loans.

Theme 2. Convenience is Key

Across the interviews, convenience was a key driver of financial choice, especially when ventures had reached the early growth phase. In almost all cases where women in the sample took high-interest consumer loans, instead of subsidized loans or business loans, their main rationale for the choice was convenience of loan processing. Speaking of why she choose a consumer loan, Karen explained:

> Now it is a bit different: more professional and faster. Now, money issues are resolved faster as everything is changing; more rivals [in banking industry] have appeared. Sometimes we must down our prices. We must take the decisions quickly. Thus, DAMU [government-subsidized] loans is too long a wait…We take only consumer loans. Quick ones. The other ones are processed for too long and we don't have time for that.

Theme 3. High Risk Aversion

Though some entrepreneurs took personal loans to fund start-up expenditure and working capital, they were always wary of high cost of capital of bank loans, displayed fear of default leading to loss of collateral, and a fear of loan denial. Even if the entrepreneurs acquired a loan, they were eager to pay it off as soon as possible as they worried about overpayment.

> I realized that if I took out a loan, then I would not be able to pay it off as the business was not yet commercial. It (the business) is unstable and initially I wasn't ready to take risks. That's why I chose personal funds… If I default—because the cashflow is unstable—then I will lose it all. I will not be able to feed my children." (Alyona)

Only three entrepreneurs (of the 14 interviewed) had experience with equity finance. They uniformly and categorically rejected investor funds because of fear of investor appropriation. Our preliminary analysis of interview data suggests a convergence between women's stated preferences for financial choice and their actual behaviors. The stated preference order to most female entrepreneurs is: (in increasing order of preference: Equity < business loans < government-subsidized loans and grants < personal consumer loans < family savings < personal savings or retained earnings.

DISCUSSION AND CONCLUSION

We set out to understand the financing choices of small business owners in Kazakhstan at various stages of the firm lifecycle—inception, early growth (capital expenditure) and maturity (working capital). So far, we have attempted to understand the choice of female entrepreneurs in the sample. Put together, the results from the quantitative and qualitative studies suggest that entrepreneurs in Kazakhstan have a higher preference for bootstrapping investment at start-up and depend on internal resources for funding capital expenditure and working capital at later phases venture life. Secondary sources of financial capital include bank loans and government-subsidized loans.

As Holmes and Kent's (1991) constrained pecking order model suggests, small firm entrepreneurs choose internal funds as initial capital source, followed by retained earnings, and finally, short-term debt in the form of bank overdrafts and/or trade credit. Our evidence does not support the constrained pecking order framework; entrepreneurs in Kazakhstan do not appear to choose short-term debt as a viable financing option at later stages of operation. The reasons, as evinced in the qualitative study, range from risk aversion to high cost of capital and onerous terms and conditions of commercial loans faced by the women entrepreneurs in Kazakhstan.

While entrepreneurs' aversion to business loans is due to high interest rates (supply gap), we find evidence for Holmes and Kent's (1991) "knowledge gap" assertion: women entrepreneurs were unable to discover apposite information to meet the conditions for accessing government-subsidized loans (such as preparation of a business plan and income statement.)

Data collection is ongoing. However, our research complements previous, but limited scholarship examining entrepreneurs' motivations and challenges in a transition economy (Bui et al., 2018). This study provides a starting point for a more detailed investigation of challenges, motivations, and constraints that male and female entrepreneurs face in a transition economy rife with institutional voids (Khanna et al., 2005).

REFERENCES

Acs, Z. J. (2010). High-impact entrepreneurship. *Handbook of Entrepreneurship Research: An Interdisciplinary Survey and Introduction*, 165–182.

Asian Development Bank. (2013). *Kazakhstan: Country Gender Assessment.* . http://www.adb.org/sites/default/files/institutional-document/34051/files/kazakhstan-country-gender-assessment.pdf

Becker-Blease, J. R., & Sohl, J. E. (2007). Do women-owned businesses have equal access to angel capital? *Journal of Business Venturing*, *22*(4), 503–521.

Berger, A. N., & Udell, G. F. (2006). A more complete conceptual framework for SME finance. *Journal of Banking & Finance*, *30*(11), 2945–2966.

Blair, M. M., & Stout, L. (1999). A team production theory of corporate law. *Virginia Law Review*, *85*(2), 247–328.

Bohr, A., Brauer, B., Gould-Davies, N., Kassenova, N., Lillis, J., Mallinson, K., Nixey, J., & Satpayev, D. (2019). *Kazakhstan: tested by transition*. Royal Institute of International Affairs.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77–101.

Bui, H. T. M., Kuan, A., & Chu, T. T. (2018). Female entrepreneurship in patriarchal society: motivation and challenges. *Journal of Small Business & Entrepreneurship*, *30*(4), 325–343.

Canning, J., Haque, M., & Wang, Y. (2012). Women at the Wheel. *New York*.

Cassar, G. (2001). The financing and capital structure of business start-ups: the importance of asset structure. *Frontiers of Entrepreneurship Research*, 252–263.

Chaudhuri, K., Sasidharan, S., & Raj, R. S. N. (2020). Gender, small firm ownership, and credit access: some insights from India. *Small Business Economics*, *54*, 1165–1181.

Churchill, N. C., & Lewis, V. L. (1983). The Five Stages of Small Business Growth. *Harvard Business Review*, *61*(3), 30–50.

Coleman, S. (2000). Access to capital and terms of credit: A comparison of men-and women-owned small businesses. *Journal of Small Business Management*, *38*(3), 37.

Eddleston, K. A., Ladge, J. J., Mitteness, C., & Balachandra, L. (2016). Do you see what I see? Signaling effects of gender and firm characteristics on financing entrepreneurial ventures. *Entrepreneurship Theory and Practice*, *40*(3), 489–514.

Elam, A. B., Brush, C. G., Greene, P. G., Baumer, B., Dean, M., Heavlow, R., & others. (2019). Global entrepreneurship monitor 2018/2019 women's entrepreneurship report. *Babson College: Smith College and the Global Entrepreneurship Research Association*.

Fabowale, L., Orser, B., & Riding, A. (1995). Gender, structural factors, and credit terms between Canadian small businesses and financial institutions. *Entrepreneurship Theory and Practice*, *19*(4), 41–65.

Filimonau, V., Matyakubov, U., Matniyozov, M., Shaken, A., & Mika, M. (2022). Women entrepreneurs in tourism in a time of a life event crisis. *Journal of Sustainable Tourism*, 1–23.

Fisher, G., Kotha, S., & Lahiri, A. (2016). Changing with the times: An integrated view of identity, legitimacy, and new venture life cycles. *Academy of Management Review*, *41*(3), 383–409. https://doi.org/10.5465/AMR.2013.0496

Holmes, S., & Kent, P. (1991). An empirical analysis of the financial structure of small and large Australian manufacturing enterprises. *Journal of Small Business Finance*, *1*(2), 141–154.

Johnson, M. A., Stevenson, R. M., & Letwin, C. R. (2018). A woman's place is in the… startup! Crowdfunder judgments, implicit bias, and the stereotype content model. *Journal of Business Venturing*, *33*(6), 813–831.

Khanna, T., Palepu, K., & Sinha, J. (2005). Strategies that fit emerging markets. *Harvard Business Review*, *83*(6), 63–80.

Konings, J., Sagyndykova, G., Subramanian, V., & Volckaert, A. (2022). The granular nature of emerging market economies: The case of Kazakhstan. *Economics of Transition and Institutional Change*.

Michel, C. (2015, March 30). *Lee Kuan Yew's Singapore as a model for Kazakhstan*. The Diplomat. https://thediplomat.com/2015/03/lee-kuan-yews-singapore-as-a-model-for-kazakhstan/

Myers, S. C. (1984). The capital structure puzzle. *The Journal of Finance*, *39*(3).

Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, *13*(2), 187–221. https://doi.org/10.1016/0304-405X(84)90023-0

Nguyen, B., & Canh, N. P. (2021). Formal and informal financing decisions of small businesses. *Small Business Economics*, *57*, 1545–1567.

Orser, B. J., Riding, A. L., & Manley, K. (2006). Women entrepreneurs and financial capital. *Entrepreneurship Theory and Practice*, *30*(5), 643–665.

Verheul, I., & Thurik, R. (2001). Start-up capital:" does gender matter?". *Small Business Economics*, *16*, 329–346.

Winborg, J., & Landström, H. (2001). Financial bootstrapping in small businesses: Examining small business managers' resource acquisition behaviors. *Journal of Business Venturing*, *16*(3), 235–254.